

ROHAIL ASIM

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SUMMARY

Software engineer and systems researcher with 5+ years building scalable, low-latency infrastructure across 5G networking, ML-driven web optimization, and full-stack product engineering. PhD from NYU with a track record of shipping production systems tested at real-world scale, from a 2M-user ed-tech platform to real-time AR/VR frameworks deployed over live 5G networks. Fluent across the stack: backend systems, cloud infrastructure (GCP, AWS, Docker), and performance-critical applications.

EDUCATION

New York University

Sep 2020 - Aug 2025

PhD, Computer Science - Advisors: Yasir Zaki, Lakshminarayanan Subramanian

Title: Designing Efficient and Equitable Networked Systems for Mobile Users in Emerging Regions

Committee: Yasir Zaki, Lakshminarayanan Subramanian, Talal Rahwan, Matteo Varvello, Anirudh Sivaraman

Fellowship: NYU Global PhD Fellowship (2020–2025)

Lahore University of Management Science (LUMS)

2015 - 2019

Bachelor of Computer Science - CGPA: 3.70/4.0 (*Graduated with Distinction*)

EXPERIENCE

Software Engineer (Full Stack)

Jan 2020 - Aug 2020

Educative Inc. Ed-tech SaaS platform, 2M+ users, team of 30 engineers

Lahore, PAK

- Led migration of the learning platform to Next.js, implementing server-side rendering (SSR) and client-side caching that reduced page load times by 50% and improved SEO across 2M+ user accounts
- Architected and shipped a company-wide Design System in TypeScript with Material-UI, cutting UI component development time by 50% for the 30-person engineering team
- Integrated Redis caching into API pipelines, improving platform throughput and reducing database load under peak traffic
- Established full-stack testing infrastructure using Jest, Selenium, and Unittest, increasing test coverage by 50% and reducing regression risk across the codebase
- Maintained and improved CI/CD pipelines (GitHub Actions) across staging and production environments, enabling reliable continuous deployment for the engineering team
- Collaborated across a large shared codebase on GitHub, contributing to frontend and backend layers with code reviews, documentation, and cross-team feature delivery

Graduate Research Engineer

Sep 2020 - Aug 2025

New York University & NYU Abu Dhabi

High-Performance Systems & ML Infrastructure

- Built **Sonic**, a distributed connectivity system using FM radio and SMS infrastructure to serve web content to devices without internet access; validated with 30 participants in a field deployment [🌐 Link](#)
- Built **Hera**, a QoE-aware framework for multi-user AR/VR over 5G, combining a custom TCP Linux kernel module in C with a WebXR frontend and backend coordination services, reducing interaction latency by 66% and improving video quality by 50% [🌐 Link](#) [🔗 Code](#)
- Designed and executed a large-scale benchmarking pipeline evaluating state-of-the-art LLMs across 32 university courses using statistical analysis (bootstrapped Welch's t-tests, OLS regression) and automated evaluation frameworks with findings covered in outlets including The Times, The Independent, Daily Mail, and Nature [📺 Media Coverage](#) [🌐 Link](#)
- Built **Zeus**, a large-scale 5G benchmarking system deployed on AWS for evaluating congestion control algorithms across multiple network conditions using Python, Mahimahi, and NS-3; the most comprehensive cross-protocol 5G CCA study to date [🌐 Link](#) [🔗 Code](#)
- Built **Lite-Web**, a production ML + rule-based system that reduced JavaScript processing by 54% and page load times by 72% on low-end devices; validated with 200 students in a real-world deployment [📺 Media Coverage](#) [🌐 Link](#)
- Implemented a multi-agent Reinforcement Learning system for autonomous vehicle routing, reducing average travel time by 30% and improving congestion fairness by 50% across simulated networks [🌐 Link](#) [🔗 Code](#)

- Built a Python-based carbon emissions calculator for AI infrastructure using GCP APIs, grid emission datasets, and datacenter energy profiles packaged with Docker for portable deployment [🌐 Link](#)

SKILLS

Languages	Python, C, C++, JavaScript, TypeScript, Go, SQL, Matlab, Haskell
Frameworks & Libraries	PyTorch, Next.js, React, Scikit-learn, Pandas, NumPy, Selenium
Cloud & Infra	GCP, AWS, Docker, Kubernetes, Redis, CI/CD (GitHub Actions), NGINX
Systems & Networking	Linux (kernel modules), TCP/IP, Wireshark, Mahimahi, NS-3, 5G/LTE
Testing	Jest, Selenium, Unittest, Large-scale empirical measurement, A/B testing

PUBLICATIONS

[1] **Tuning into the Web: A Low-Cost Access Solution for Developing Countries**
Ayush Pandey, **Rohail Asim**, Jean Louis K. E. Fendji, Talal Rahwan, Matteo Varvello, Yasir Zaki
Accepted at ACM SIGCOMM 2026

[2] **Self-Regulating Cars: Automating Traffic Control in Free-Flow Road Networks**
Ankit Bhardwaj, **Rohail Asim**, Kevin Jin, Yasir Zaki, Lakshminarayanan Subramanian
AAAI AISI 2026

[3] **Modeling Economic Viability for Scalable AI Deployment in Emerging Regions**
Rohail Asim, Ankit Bhardwaj, Arjuna Sathiaselam, Yasir Zaki, and Lakshminarayanan Subramanian
ACM SOSP PACMI 2025

[4] **The GAIUS Experience: Powering a Hyperlocal Mobile Web for Communities in Emerging Regions**
Rohail Asim, Arjuna Sathiaselam, Arko Chatterjee, Mukund Lal, Yasir Zaki, Lakshminarayanan Subramanian
ACM ICTD 2024

[5] **SONIC: Connect the Unconnected via FM Radio & SMS**
Ayush Pandey, **Rohail Asim**, Khalid Mengal, Matteo Varvello, and Yasir Zaki
ACM CoNEXT 2024

[6] **Impact of Congestion Control on Mixed Reality Applications**
Rohail Asim, Lakshminarayanan Subramanian, and Yasir Zaki
ACM SIGCOMM EMS 2024

[7] **I tag, you tag, everybody tags!**
Hazem Ibrahim, **Rohail Asim**, Matteo Varvello, Yasir Zaki
ACM Internet Measurement Conference (IMC) 2023

[8] **Perception, performance, and detectability of conversational artificial intelligence across 32 university courses**
Hazem Ibrahim, Fengyuan Liu, **Rohail Asim**, Yasir Zaki et al
Scientific Reports 2023

[9] **Rethinking homework in the age of artificial intelligence**
Hazem Ibrahim, **Rohail Asim**, Fareed Zaffar, Talal Rahwan, Yasir Zaki
IEEE Intelligent Systems

[10] **Towards a world wide web without digital inequality**
M Chaqfeh, **Rohail Asim**, Bedoor AlShebli, Fareed Zaffar, Talal Rahwan, Yasir Zaki
Proceedings of the National Academy of Sciences (PNAS)

[11] **ALCC: Migrating Congestion Control To The Application Layer In Cellular Networks**
Yasir Zaki, **Rohail Asim**, Muhammad Khan, Shiva Iyer, Talal Ahmad, Thomas Potsch, Lakshminarayanan Subramanian
Journal of Systems Research

Full publication list at [Google Scholar](#)

HONORS & AWARDS

NYU Global PhD Fellowship	2020 – 2025
Dean’s Honor List	Academic Years 2017–2018, 2018–2019
Graduated with Distinction	2019